

Exercise Sheet 1

Due date: 15:30, Oct 6th.
Late submissions will receive no credit.

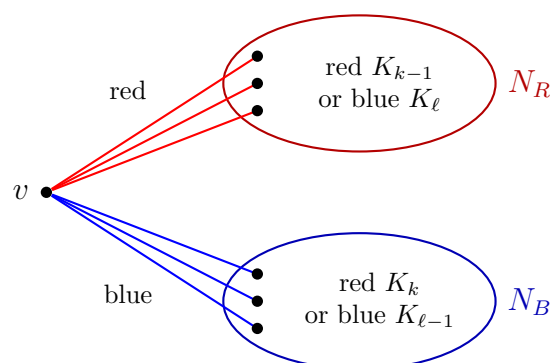
You and your partner should try to solve all of the exercises below, and then submit a total of two solutions to be graded — each problem is worth 10 points. Both partners should author one solution each, making clear who wrote which solution, but you will earn credit for both solutions. Make sure to clearly justify your answers, defining everything precisely and stating any results you are using. You should then submit your solutions on NTU COOL, either as a typed PDF or a clear picture or scan of your handwritten answers.

Exercise 1 Given natural numbers k, ℓ , we define the asymmetric Ramsey number $R(k, \ell)$ as the smallest n such that every red-/blue-colouring of the edges of K_n contains a red K_k or a blue K_ℓ .

- (a) Show that for $k, \ell \geq 2$, we have $R(k, \ell) \leq R(k, \ell - 1) + R(k - 1, \ell)$.
- (b) Deduce that for $k, \ell \geq 1$, we have $R(k, \ell) \leq \binom{k+\ell-2}{k-1}$.
- (c) Sharpen this bound to show $R(3, 4) \leq 9$.

Solution. Throughout, note that $R(1, \ell) = R(k, 1) = 1$, since K_1 has no edges and hence is trivially both a red K_1 and a blue K_1 .

Proof of (a). If $R(k, \ell - 1)$ or $R(k - 1, \ell)$ is infinite there is nothing to prove, so assume both are finite and let $n = R(k, \ell - 1) + R(k - 1, \ell)$. Take an arbitrary red-/blue-colouring of the edges of K_n and fix a vertex v . Let N_R (resp. N_B) be the set of vertices joined to v by a red (resp. blue) edge. Then $|N_R| + |N_B| = n - 1$, so either $|N_R| \geq R(k - 1, \ell)$ or $|N_B| \geq R(k, \ell - 1)$; otherwise $n - 1 \leq R(k - 1, \ell) + R(k, \ell - 1) - 2 = n - 2$, a contradiction.



Case 1: $|N_R| \geq R(k-1, \ell)$. The colouring restricted to the complete graph on N_R contains a red K_{k-1} or a blue K_ℓ . A blue K_ℓ is what we want. If it contains a red K_{k-1} , then adding v , which is joined to all of its vertices by red edges, gives a red K_k .

Case 2: $|N_B| \geq R(k, \ell-1)$. Symmetrically, the complete graph on N_B contains a red K_k , or a blue $K_{\ell-1}$, which together with v forms a blue K_ℓ .

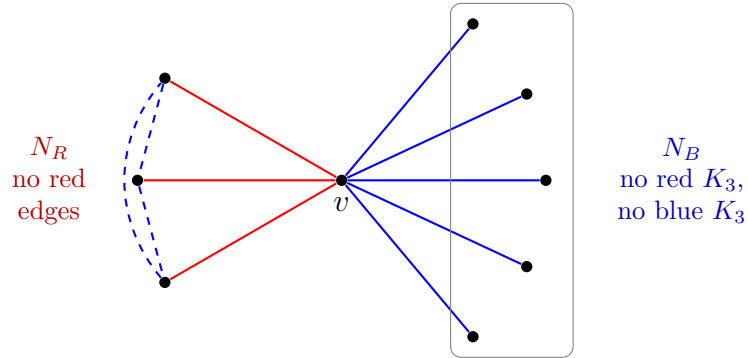
Hence every colouring of K_n contains a red K_k or a blue K_ℓ , so $R(k, \ell) \leq n = R(k, \ell-1) + R(k-1, \ell)$. $\square$

Proof of (b). We induct on $k + \ell$. If $k = 1$ or $\ell = 1$, then $R(k, \ell) = 1 = \binom{k+\ell-2}{k-1}$, since $\binom{\ell-1}{0} = \binom{k-1}{k-1} = 1$. Now let $k, \ell \geq 2$ and assume the claim for all pairs with smaller sum. In particular $R(k, \ell-1)$ and $R(k-1, \ell)$ are finite, and by (a) and Pascal's rule,

$$R(k, \ell) \leq R(k, \ell-1) + R(k-1, \ell) \leq \binom{k+\ell-3}{k-1} + \binom{k+\ell-3}{k-2} = \binom{k+\ell-2}{k-1}.$$

$\square$

Proof of (c). By (b), $R(3, 3) \leq \binom{4}{2} = 6$. Suppose, for contradiction, that some colouring of K_9 has neither a red K_3 nor a blue K_4 . Fix a vertex v and define N_R, N_B as in (a), so $|N_R| + |N_B| = 8$.



- $|N_R| \leq 3$: if $x, y \in N_R$ were joined by a red edge, then $\{v, x, y\}$ would be a red K_3 . So all edges inside N_R are blue, and if $|N_R| \geq 4$, any four vertices of N_R form a blue K_4 .
- $|N_B| \leq 5$: if $|N_B| \geq 6 \geq R(3, 3)$, then N_B contains a red K_3 , or a blue K_3 , which together with v forms a blue K_4 .

Thus $|N_R| = 8 - |N_B| \geq 3$, so $|N_R| = 3$. That is, every vertex is incident to exactly 3 red edges, so the red edges form a 3-regular graph on 9 vertices. But by the handshake lemma the sum of degrees, $9 \cdot 3 = 27$, must equal twice the number of red edges, which is impossible since 27 is odd. Hence every colouring of K_9 contains a red K_3 or a blue K_4 , i.e. $R(3, 4) \leq 9$. $\square$

Exercise 2 Let H be a k -uniform hypergraph on n vertices V with edges $E = \{e_1, e_2, \dots, e_m\}$. To each vertex $v \in V$, associate a uniformly random number $x(v) \in [0, 1]$. Order the vertices in increasing order of the numbers $x(v)$, and colour them greedily — colour the vertex red, unless it would create an all-red edge, in which case colour it blue.

Fix some $\varepsilon > 0$, and consider the following events:

- E_i : the event that for every vertex v in the edge e_i , we have $x(v) \leq \frac{1}{2}(1 - \varepsilon)$.
- L_i : the event that for every vertex v in the edge e_i , we have $x(v) \geq \frac{1}{2}(1 + \varepsilon)$.
- $M_{i,j}$: the event that edges e_i and e_j have a single common vertex w , with $\frac{1}{2}(1 - \varepsilon) < x(w) < \frac{1}{2}(1 + \varepsilon)$, and that for every vertex $u \in e_i$ we have $x(u) \leq x(w)$, and for every vertex $v \in e_j$ we have $x(v) \geq x(w)$.

- (a) Show that H has Property B (i.e. is two-colourable), unless the event $(\cup_{i \in [m]} E_i) \cup (\cup_{i \in [m]} L_i) \cup (\cup_{i,j \in [m]} M_{i,j})$ occurs.
- (b) Bound the probability of this event in terms of m and ε .
- (c) By choosing ε appropriately, show that $m(k) \geq c \left(\frac{k}{\log k} \right)^{1/2} 2^k$, for some constant $c > 0$.

Exercise 3 Show that there is some constant $c > 0$ such that, if n is sufficiently large and every vertex of an n -vertex graph G has degree at least $\log n - c \log \log n$, then we can colour the vertices of G red and blue in such a way that every vertex has both a red neighbour and a blue neighbour.

Exercise 4 Prove that there is some constant $c > 0$ such that if a tournament has the k -Schütte property, it must have at least $ck2^k$ vertices.

Exercise 5 We say a k -uniform hypergraph is rainbow- r -colourable if its vertices can be r -coloured such that every edge sees all r colours.

- (a) Prove that if a k -uniform hypergraph H has at most $\frac{r^{k-1}}{(r-1)^k}$ edges, then it is rainbow- r -colourable.
- (b) By considering a random hypergraph, or otherwise, show that there exists, for some constant $C = C(r) > 0$ and arbitrarily large k , a k -uniform hypergraph H with $Ck^2 \left(\frac{r}{r-1} \right)^k$ edges that is not rainbow- r -colourable.